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Research paper

Controllability and observability of tempered fractional differential systems

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ABSTRACT

We study controllability and observability concepts of tempered fractional linear systems in the Caputo sense. First, we formulate a solution for the class of tempered systems under investigation by means of the Laplace transform method. Then, we derive necessary and sufficient conditions for the controllability, as well as for the observability, in terms of the Gramian controllability matrix and the Gramian observability matrix, respectively. Moreover, we establish the Kalman criteria that allows one to check easily the controllability and the observability for tempered fractional systems. Applications to the fractional Chua's circuit and Chua–Hartley's oscillator models are provided to illustrate the theoretical results developed in this manuscript.

1. Introduction

Tempered fractional calculus extends the traditional framework of fractional calculus by introducing a tempering parameter that controls the decay rate of the memory kernel [1,2]. This extension allows for more flexible modeling of systems with long-range memory effects and decay, capturing a wider range of behaviors beyond those described by classical fractional operators, such as viscoelastic materials, anomalous diffusion, and systems where, after a long period, the influence of the past decays at a certain rate. In recent years, tempered fractional calculus has become a subject of investigation due to its applications in stochastic and dynamical systems [3–6]. For example, it has been used to describe and understand turbulence in geophysical flows [3] and Lévy processes such as Brownian motion [3,6]. Additionally, a truncated Lévy flight has been investigated to capture the natural cutoff in real physical systems [4]. Furthermore, stochastic tempered Lévy flights have led to significant advancements in mathematical research, including solving multi-dimensional partial differential equations using both analytical and numerical methods [7].

In control theory, the controllability and observability concepts are fundamental to the study of dynamical systems. Controllability enables one to steer the system's state from an initial state to a desired state using control inputs, while observability inform us about the ability to reconstruct the system's internal state based on its outputs [8]. Matingon and d'Andréa–Novel were the first to have examined the controllability and observability properties of linear fractional differential equations in finite dimensional spaces, either in state space form or in polynomial representation [9]. Then, several authors have been investigating various results about the controllability and observability of linear and nonlinear fractional differential equations, see for example [10–14] and references

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cited therein. For instance, in [10], the observability and controllability of a fractional time invariant linear system is investigated by means of the observability and controllability Grammian matrices that are obtained in terms of the Mittag-Leffler matrix function. Also, in [14], necessary and sufficient conditions of controllability and observability of fractional continuous time linear systems with regular pencils are derived via the construction of Gramian matrices. In [15], Balachandran et al. studied the controllability of linear and nonlinear fractional damped dynamical systems involving fractional Caputo derivatives of varying orders in finite-dimensional spaces. Their approach utilized the Mittag-Leffler matrix function and an iterative technique. Furthermore, in [12], Matar presented some properties of controllability and observability of fractional linear systems using conformable fractional derivatives, where he proved that the controllability is equivalent to a controllability matrix that has a full rank and established a relationship between the controllability and the fractional differential Lyapunov equation.

For tempered fractional systems, we have only some results about the stability and stabilizability [16–19]. For instance, in [16], the Mittag-Leffler stability of tempered dynamical systems is studied based on the Lyapunov direct method and the fractional comparison principle. Moreover, in [17], the generalized practical Mittag-Leffler stability of a class of tempered fractional nonlinear systems is investigated. In [18], the asymptotic and Mittag-Leffler stability of tempered fractional neural networks are analyzed by means of the Banach fixed point theorem, while in [19], a time delay dependent and independent criteria for the finite time stability of tempered fractional systems with delay are characterized.

Motivated by the proceeding, in this work we propose to investigate the controllability and observability problems of tempered fractional linear systems, which have not yet been tackled. The main contributions of this paper can be summarized as follows:

- An analytical solution to tempered linear systems is obtained by means of the Laplace transform method (see Theorem 3.1).
- Necessary and sufficient conditions are established for the controllability and observability problems by means of Gramian matrices (see Theorems 3.2 and 3.4, respectively).
- Kalman criteria are derived for the controllability and observability of a class of tempered linear systems (see Theorems 3.3 and 3.5, respectively).
- The obtained controllability and observability results are applied to the tempered fractional Chua's circuit model (see Section 4.1) and Chua–Hartley's oscillator model (see Section 4.2).

The remainder of this article is organized as follows. In Section 2, we provide the necessary preliminaries, including the definitions of tempered fractional derivative and integral in the Caputo sense. Afterwards, we proceed with Section 3, where we begin by proving an important result about the Laplace transform of two functions that depend on Mittag-Leffler functions of one and two parameters. Then, we establish the analytical solution for tempered fractional linear systems. Furthermore, we state our main controllability and observability results. Applications that illustrate the obtained results are given in Section 4. We end up with Section 5 of conclusion and some possible directions of future research.

2. Preliminaries

In this section, we state some basic definitions and preliminary results about tempered fractional operators, which are used in the rest of the paper.

Definition 2.1 (*See [1,2,20]*). Let $\alpha > 0$, $\rho > 0$, and υ be an absolutely integrable function defined on $[a, b]$, $a, b \in \mathbb{R}$, and $a < b$ (if $b = \infty$, then the interval is half-open). The tempered fractional integral of function υ is defined as follows:

$$ {}^{T}I_{a}^{\alpha,\rho}\upsilon(t) = \frac{1}{\Gamma(\alpha)}\int_{a}^{t} e^{-\rho(t-s)}(t-s)^{\alpha-1}\upsilon(s)\mathrm{d}s, \tag{1}$$

where $\Gamma(\cdot)$ represents the Euler gamma function defined by

$$\Gamma(z) = \int_{0}^{\infty} e^{-s}s^{z-1}\,\mathrm{d}s, \quad z \in \mathbb{C}.$$

Definition 2.2 (*See [1,2]*). Let $\alpha \in (0,1)$ and $\rho > 0$. The Caputo tempered fractional order derivative of a function $\upsilon \in C^{1}([0,b],\mathbb{R})$ is given by

$$ {}^{T}D_{a}^{\alpha,\rho}\upsilon(t) = \frac{1}{\Gamma(1-\alpha)}\int_{a}^{t} e^{-\rho(t-s)}(t-s)^{-\alpha}D^{1,\rho}\upsilon(s)\mathrm{d}s \tag{2}$$

with $D^{1,\rho}\upsilon(t) = \rho\upsilon(t) + d\upsilon'(t)$, where $d = 1$ and its dimension equals the dimension of the independent variable t.

Remark 2.1. To ensure the dimensional homogeneity within the combination $\rho\upsilon(t) + d\upsilon'(t)$, the constant $d = 1$ with a dimension equal to the dimension of the independent variable t is added [19].

Remark 2.2. If we let $\rho = 0$ in Definition 2.2, then we retrieve the classical fractional derivative in Caputo sense [21].

Definition 2.3 (*See [22]*). The Mittag-Leffler functions of a matrix A, of one and two parameters, are defined as

$$E_{\alpha}(A) = \sum_{l=0}^{+\infty} \frac{A^{l}}{\Gamma(\alpha l + 1)}, \quad Re(\alpha) > 0, \tag{3}$$

and

$$E_{\alpha,\alpha'}(A)=\sum_{l=0}^{+\infty}\frac{A^l}{\Gamma(\alpha l+\alpha')},\quad Re(\alpha)>0,\ \alpha'>0, \tag{4}$$

respectively.

Lemma 2.1 (*See [23]*). *The Laplace transform of a function $\varphi(t)$ of a real variable $t\in\mathbb{R}^+$ is defined by*

$$(\mathcal{L}\varphi)(s)=\mathcal{L}[\varphi(t)](s)=\bar{\varphi}(s):=\int_0^\infty e^{-st}\,\varphi(t)dt,\quad s\in\mathbb{C}.$$

Lemma 2.2 (*See [1]*). *The Laplace transform of the tempered fractional integral (1) and the Caputo derivative (2) are given as*

1. $\mathcal{L}\left[{}^T I_0^{\alpha,\rho} v(t)\right](s)=(\rho+s)^{-\alpha}\mathcal{L}[v(t)](s)$;
2. $\mathcal{L}\left[{}^T D_0^{\alpha,\rho} v(t)\right](s)=(s+\rho)^{\alpha}\mathcal{L}[v(t)](s)-(s+\rho)^{\alpha-1}v(0)$.

3. Main results

In this paper, we are interested in studying the analytical solution, the controllability, as well as the observability, of the following linear tempered fractional system:

$$\begin{cases} {}^T D_0^{\alpha,\rho} y(t)=Ay(t)+Bu(t), & t\in[0,T],\\ y(0)=y_0, & y_0\in\mathbb{R}^n,\end{cases} \tag{5}$$

where $y:[0,T]\to\mathbb{R}^n$ is the state, ${}^T D_0^{\alpha,\rho}y(t)$ is the Caputo tempered fractional order derivative of y, y_0 is the initial state, $A\in\mathbb{R}^{n\times n}$ is the state matrix, $B\in\mathbb{R}^{n\times m}$ is the control matrix, and $u:[0,T]\to\mathbb{R}^m$ is the input control function.

We first prove the following fundamental lemma that enables us to establish Theorem 3.1.

Lemma 3.1. *Let $\alpha\in(0,1)$. Then,*

1. $\mathcal{L}\left[e^{-\rho t}E_\alpha(At^\alpha)\right](s)=(s+\rho)^{\alpha-1}\left((s+\rho)^\alpha I-A\right)^{-1}$;
2. $\mathcal{L}\left[e^{-\rho t}t^{\alpha-1}E_{\alpha,\alpha}(At^\alpha)\right](s)=[(s+\rho)^\alpha I-A]^{-1}$.

Proof. Using the fact that $\mathcal{L}\left[e^{ct}f(t)\right](s)=\mathcal{L}[f(t)](s-c)$, we have

$$\mathcal{L}\left[E_\alpha(At^\alpha)\right](s+\rho)=\mathcal{L}\left[e^{-\rho t}E_\alpha(At^\alpha)\right](s) \tag{6}$$

and

$$\mathcal{L}\left[t^{\alpha-1}E_{\alpha,\alpha}(At^\alpha)\right](s+\rho)=\mathcal{L}\left[e^{-\rho t}t^{\alpha-1}E_{\alpha,\alpha}(At^\alpha)\right](s). \tag{7}$$

On the other hand, from Definition 2.3, one has

$$\begin{aligned}\mathcal{L}\left[E_\alpha(At^\alpha)\right](s+\rho)&=\mathcal{L}\left[\sum_{k=0}^{\infty}\frac{(At^\alpha)^k}{\Gamma(k\alpha+1)}\right](s+\rho)\\ &=\sum_{k=0}^{\infty}\frac{A^k}{\Gamma(k\alpha+1)}\mathcal{L}[t^{\alpha k}](s+\rho)\\ &=\sum_{k=0}^{\infty}\frac{A^k}{\Gamma(k\alpha+1)}\frac{\Gamma(\alpha k+1)}{(s+\rho)^{\alpha k+1}}.\end{aligned}$$

Using $\mathcal{L}[t^{\alpha k}](s+\rho)=\dfrac{\Gamma(\alpha+1)}{(s+\rho)^{\alpha+1}}$ yields

$$\mathcal{L}\left[E_\alpha(At^\alpha)\right](s+\rho)=\frac{1}{(s+\rho)}\sum_{k=0}^{\infty}\left(\frac{A}{(s+\rho)^\alpha}\right)^k. \tag{8}$$

Without loss of generality, let us assume that $s\geq\|A\|^{\frac{1}{\alpha}}$. Then, $\|A\|^{\frac{1}{\alpha}}\leq(s+\rho)$ and

$$\left\|\frac{A}{(s+\rho)^\alpha}\right\|<1. \tag{9}$$

Applying (9) to (8), one gets that

$$\mathcal{L}\left[E_\alpha(At^\alpha)\right](s+\rho)=(s+\rho)^{\alpha-1}\left((s+\rho)^\alpha I-A\right)^{-1}. \tag{10}$$

Hence, by combining (6) and (10), we obtain that

$$\mathcal{L}\left[e^{-\rho t}E_\alpha(At^\alpha)\right](s)=(s+\rho)^{\alpha-1}\left((s+\rho)^\alpha I-A\right)^{-1}.$$

This proves the first statement of the lemma. For the second statement, we have

$$\begin{aligned}\mathcal{L}\left[t^{\alpha-1}E_{\alpha,\alpha}\left(At^{\alpha}\right)\right](s+\rho) &= \mathcal{L}\left[t^{\alpha-1}\sum_{k=0}^{\infty}\frac{(At^{\alpha})^{k}}{\Gamma(\alpha k+\alpha)}\right](s+\rho)\\ &= \mathcal{L}\left[\sum_{k=0}^{\infty}\frac{t^{\alpha k+\alpha-1}A^{k}}{\Gamma(\alpha k+\alpha)}\right](s+\rho)\\ &= \sum_{k=0}^{\infty}\frac{A^{k}}{\Gamma(\alpha k+\alpha)}\mathcal{L}[t^{\alpha k+\alpha-1}](s+\rho)\\ &= \sum_{k=0}^{\infty}\frac{A^{k}}{\Gamma(\alpha k+\alpha)}\frac{\Gamma(\alpha k+\alpha)}{(s+\rho)^{\alpha k+\alpha}}.\end{aligned}$$

Therefore,

$$\mathcal{L}\left[t^{\alpha-1}E_{\alpha,\alpha}\left(At^{\alpha}\right)\right](s+\rho) = \left((s+\rho)^{\alpha}I-A\right)^{-1}. \tag{11}$$

By combining (7) and (11), it follows that

$$\mathcal{L}\left[\mathrm{e}^{-\rho t}\,t^{\alpha-1}E_{\alpha,\alpha}\left(At^{\alpha}\right)\right](s) = \left((s+\rho)^{\alpha}I-A\right)^{-1}.$$

The proof is complete. □

The following theorem characterizes the well-posedness of system (5).

Theorem 3.1. *Let $\rho>0$ and $\alpha\in(0,1)$. Consider $u(t)\in\mathbb{R}^{m}$, $A\in\mathbb{R}^{n\times n}$, and $B\in\mathbb{R}^{n\times m}$, with the property that $\det[(s+\rho)^{\alpha}I-A]\neq 0$ for $s\geq\|A\|^{\frac{1}{\alpha}}$. Then, the solution of the linear tempered fractional system (5) is given by*

$$y(t) = \mathrm{e}^{-\rho t}\,E_{\alpha}\left(At^{\alpha}\right)y_0 + \int_0^t \mathrm{e}^{-\rho(t-\tau)}(t-\tau)^{\alpha-1}E_{\alpha,\alpha}\left(A(t-\tau)^{\alpha}\right)Bu(\tau)d\tau. \tag{12}$$

Proof. Applying the Laplace transform to both sides of the first equation of system (5), we obtain that

$$\mathcal{L}\left[{}^{T}D_0^{\alpha,\rho}y(t)\right](s) = \mathcal{L}\left[Ay(t)+Bu(t)\right](s). \tag{13}$$

Using the linearity of the Laplace transform, Eq. (13) implies that

$$\begin{aligned}\mathcal{L}\left[{}^{T}D_0^{\alpha,\rho}y(t)\right](s) &= \mathcal{L}\left[Ay(t)\right](s)+\mathcal{L}\left[Bu(t)\right](s)\\ &= A\mathcal{L}[y(t)](s)+B\mathcal{L}[u(t)](s).\end{aligned} \tag{14}$$

Eq. (14) is obtained due to the definition of $\mathcal{L}$. Specifically,

$$\mathcal{L}\left[Ay(t)\right](s) = \int_0^{\infty}\mathrm{e}^{-st}\,Ay(t)dt = A\int_0^{\infty}\mathrm{e}^{-st}\,y(t)dt$$

and

$$\mathcal{L}\left[Ay(t)\right](s) = \int_0^{\infty}\mathrm{e}^{-st}\,By(t)dt = B\int_0^{\infty}\mathrm{e}^{-st}\,y(t)dt.$$

Applying Lemma 2.2, we obtain from Eq. (14) that

$$(s+\rho)^{\alpha}\mathcal{L}[y(t)](s)-(s+\rho)^{\alpha-1}y_0 = A\mathcal{L}[y(t)](s)+B\mathcal{L}[u(t)](s).$$

Thus,

$$\left((s+\rho)^{\alpha}I-A\right)\mathcal{L}[y(t)](s) = (s+\rho)^{\alpha-1}y_0+B\mathcal{L}[u(t)](s).$$

Since $((s+\rho)^{\alpha}I-A)$ is invertible, we have

$$\mathcal{L}[y(t)](s) = \left((s+\rho)^{\alpha}I-A\right)^{-1}(s+\rho)^{\alpha-1}y_0+\left((s+\rho)^{\alpha}I-A\right)^{-1}B\mathcal{L}[u(t)](s). \tag{15}$$

By virtue of Lemma 3.1, it follows that

$$\begin{aligned}\mathcal{L}[y(t)](s) &= \mathcal{L}\left[\mathrm{e}^{-\rho t}\,E_{\alpha}\left(At^{\alpha}\right)\right](s)y_0+\mathcal{L}\left[\mathrm{e}^{-\rho t}\,t^{\alpha-1}E_{\alpha,\alpha}\left(At^{\alpha}\right)\right](s)B\mathcal{L}[u(t)](s)\\ &= \mathcal{L}\left[\mathrm{e}^{-\rho t}\,E_{\alpha}\left(At^{\alpha}\right)\right](s)y_0+\mathcal{L}\left[\mathrm{e}^{-\rho t}\,t^{\alpha-1}E_{\alpha,\alpha}\left(At^{\alpha}\right)*Bu(t)\right](s),\end{aligned}$$

where $*$ denotes the convolution integral. Taking the inverse Laplace transform of the above equation, we obtain that

$$y(t) = \mathrm{e}^{-\rho t}\,E_{\alpha}\left(At^{\alpha}\right)y_0+\mathrm{e}^{-\rho t}\,t^{\alpha-1}*E_{\alpha,\alpha}\left(At^{\alpha}\right)Bu(t).$$

Then, from the definition of convolution, we get

$$y(t) = \mathrm{e}^{-\rho t}\,E_{\alpha}\left(At^{\alpha}\right)y_0+\int_0^t\mathrm{e}^{-\rho(t-\tau)}(t-\tau)^{\alpha-1}E_{\alpha,\alpha}\left(A(t-\tau)^{\alpha}\right)Bu(\tau)d\tau,$$

which proves the intended result. □

3.1. Controllability analysis

In this part, we present some results concerning the controllability of system (5). Recalling Theorem 3.1, the solution of system (5) satisfies the following variation of constants formula:

$$y(t) = \mathrm{e}^{-\rho t}\, E_\alpha\left(At^\alpha\right) y_0 + \int_0^t \mathrm{e}^{-\rho(t-\tau)}(t-\tau)^{\alpha-1} E_{\alpha,\alpha}\left(A(t-\tau)^\alpha\right) Bu(\tau)d\tau. \tag{16}$$

Definition 3.1. System (5) is said to be state controllable on $[0,T]$, where $T>0$, if there exists an input signal $u(\cdot):[0,T]\to\mathbb{R}^m$ such that the associated solution of (5) satisfies $y(T)=0$.

Definition 3.1 concerns controllability at the origin, which is equivalent to controllability at any other state due to the linearity of the system.

Definition 3.2. System (5) is controllable if for some $T>0$, the controllability map $L_T : L^2([0,T];\mathbb{R}^m) \longrightarrow \mathbb{R}^n$ has $\mathbb{R}^n$ as range, where

$$L_T u := \int_0^T \mathrm{e}^{-\rho(T-\tau)}(T-\tau)^{\alpha-1} E_{\alpha,\alpha}\left(A(T-\tau)^\alpha\right) Bu(\tau)d\tau.$$

Remark 3.1. Definition 3.2 is an extension of the alternative controllability definition for integer order differential linear systems, as given in [8].

A criterion that can be used to construct an open-loop control signal consists of introducing the controllability Gramian matrix given by

$$W_c[0,T] := L_T L_T^* = \int_0^T \mathrm{e}^{-2\rho(T-\tau)}(T-\tau)^{2\alpha-2} E_{\alpha,\alpha}\left((T-\tau)^\alpha A\right) BB^* E_{\alpha,\alpha}\left((T-\tau)^\alpha A^*\right) \mathrm{d}\tau, \tag{17}$$

where we denote the matrix transpose by *.

Theorem 3.2. *Under the hypotheses of Theorem 3.1, system (5) is controllable on $[0,T]$ if, and only if, $W_c[0,T]$ is non-singular for all $T>0$.*

Proof. Let $T>0$ and suppose that $W_c[0,T]$ is non-singular. We show that $y(T)=0$. Since $W_c[0,T]$ is non-singular, $W_c^{-1}[0,T]$ is well-defined. On one hand, replacing t by T in formula (16), we obtain that

$$y(T) = \mathrm{e}^{-\rho T}\, E_\alpha\left(AT^\alpha\right) y_0 + \int_0^T \mathrm{e}^{-\rho(T-\tau)}(T-\tau)^{\alpha-1} E_{\alpha,\alpha}\left(A(T-\tau)^\alpha\right) Bu(\tau)d\tau. \tag{18}$$

The control sought is given in the form

$$u(\tau) := -(T-\tau)^{\alpha-1}\mathrm{e}^{-\rho(T-\tau)}(B)^* E_{\alpha,\alpha}\left(A^*(T-\tau)^\alpha\right) \times W_c^{-1}[0,T]\,\mathrm{e}^{-\rho T} \times E_\alpha\left(T^\alpha A\right) y_0. \tag{19}$$

By replacing (19) in (18),

$$\begin{aligned} y(T) &= \mathrm{e}^{-\rho T}\, E_\alpha\left(AT^\alpha\right) y_0 + \int_0^T \mathrm{e}^{-\rho(T-\tau)}(T-\tau)^{\alpha-1} E_{\alpha,\alpha}\left(A(T-\tau)^\alpha\right) \\ &\qquad B\left\{-(B)^* E_{\alpha,\alpha}^*\left(A(T-\tau)^\alpha\right) \times W_c^{-1}[0,T]\,\mathrm{e}^{-\rho T} \times E_\alpha\left(T^\alpha A\right) y_0\right\} d\tau \\ &= \mathrm{e}^{-\rho T}\, E_\alpha\left(AT^\alpha\right) y_0 - W_c[0,T]\, W_c^{-1}[0,T]\,\mathrm{e}^{-\rho T}\, E_\alpha\left(AT^\alpha\right) y_0 \\ &= \mathrm{e}^{-\rho T}\, E_\alpha\left(AT^\alpha\right) y_0 - \mathrm{e}^{-\rho T}\, E_\alpha\left(AT^\alpha\right) y_0 \\ &= 0. \end{aligned}$$

Conversely, let us assume that the system is controllable and show that the Gramian matrix is invertible. For the sake of argument, we suppose that the Gramian matrix is not invertible, that is, $\det(A)=0$. Then there exists a nonzero vector $x\in\mathbb{R}^n$ with $x^* W_c[0,T]x = 0$. Thus,

$$\begin{aligned} & x^*\left\{\int_0^T \mathrm{e}^{-2\rho(T-\tau)}(T-\tau)^{2\alpha-2} E_{\alpha,\alpha}\left((T-\tau)^\alpha A\right) BB^* E_{\alpha,\alpha}^*\left((T-\tau)^\alpha A\right) \mathrm{d}\tau\right\} x \\ &= \int_0^T x^*\, \mathrm{e}^{-2\rho(T-\tau)}(T-\tau)^{2\alpha-2} E_{\alpha,\alpha}\left((T-\tau)^\alpha A\right) BB^* E_{\alpha,\alpha}^*\left((T-\tau)^\alpha A\right) x\mathrm{d}\tau \\ &= 0, \end{aligned}$$

which implies that

$$\int_0^T \mathrm{e}^{-2\rho(T-\tau)}(T-\tau)^{2\alpha-2}\| y^* E_{\alpha,\alpha}\left((T-\tau)^\alpha A\right) B\|^2 d\tau = 0.$$

Since $(T-\tau)^{2\alpha-2} > 0$ for all $\tau \in [0,T]$, then $e^{-2\rho(T-\tau)}(T-\tau)^{2\alpha-2} > 0$ and

$$\|y^* E_{\alpha,\alpha}((T-\tau)^\alpha A) B\|^2 = 0.$$

Therefore,

$$y^* E_{\alpha,\alpha}((T-\tau)^\alpha A) B = 0. \tag{20}$$

Let $y_0 = E_\alpha^{-1}(T^\alpha A)x$. By the supposition of controllability, there exists a control u with the property that it moves y_0 to the origin. Then, for $y_0 = E_\alpha^{-1}(T^\alpha A)x$, formula (16) implies that

$$\begin{aligned} y(T) &= e^{-\rho T} E_\alpha(AT^\alpha) E_\alpha^{-1}(T^\alpha A)x + \int_0^T e^{-\rho(T-\tau)}(T-\tau)^{\alpha-1} E_{\alpha,\alpha}(A(T-\tau)^\alpha) Bu(\tau)d\tau \\ &= e^{-\rho T} x + \int_0^T e^{-\rho(T-\tau)}(T-\tau)^{\alpha-1} E_{\alpha,\alpha}(A(T-\tau)^\alpha) Bu(\tau)d\tau \\ &= 0. \end{aligned}$$

By taking the scalar product of the equation above with x, we obtain

$$x^* e^{-\rho T} x + \int_0^T x^* e^{-\rho(T-\tau)}(T-\tau)^{\alpha-1} E_{\alpha,\alpha}(A(T-\tau)^\alpha) Bu(\tau)d\tau = 0.$$

Thus,

$$e^{-\rho T} x^* x + \int_0^T e^{-\rho(T-\tau)}(T-\tau)^{\alpha-1} x^* E_{\alpha,\alpha}(A(T-\tau)^\alpha) Bu(\tau)d\tau = 0.$$

It follows from (20) that

$$e^{-\rho T} \|x\|^2 = 0.$$

This gives $x = 0$, which is absurd with x a nonzero vector. Hence, $W_c[0,T]$ is invertible. □

The following result presents a simple algebraic criterion to check the controllability of system (5). It is an extension of the well-known Kalman criterion for controllability.

Theorem 3.3. *Under the hypotheses of Theorem 3.1, the tempered fractional system (5) is controllable if, and only if, the Kalman matrix*

$$K = \left[B, AB, A^2B, \dots, A^{n-1}B\right]$$

has full rank.

Proof. We have

$$\begin{aligned} y(T) - e^{-\rho T} E_\alpha(AT^\alpha) y_0 &= \int_0^T e^{-\rho(T-\tau)}(T-\tau)^{\alpha-1} E_{\alpha,\alpha}(A(T-\tau)^\alpha) Bu(\tau)d\tau \\ &= \int_0^T e^{-\rho(T-\tau)}(T-\tau)^{\alpha-1} \sum_{k=0}^{\infty} \frac{[(T-\tau)^\alpha A]^k}{\Gamma(\alpha k+\alpha)} Bu(\tau)d\tau. \end{aligned}$$

By using the uniform convergence, it follows that

$$\begin{aligned} y(T) - e^{-\rho T} E_\alpha(AT^\alpha) y_0 &= \sum_{k=0}^{\infty} \int_0^T e^{-\rho(T-\tau)}(T-\tau)^{\alpha-1} \frac{(T-\tau)^\alpha A^k}{\Gamma(\alpha k+\alpha)} Bu(\tau)d\tau \\ &= \sum_{k=0}^{\infty} A^k B \int_0^T e^{-\rho(T-\tau)}(T-\tau)^{\alpha-1} \frac{(T-\tau)^\alpha}{\Gamma(\alpha k+\alpha)} u(\tau)d\tau \\ &= \lim_{N\to\infty} \sum_{k=0}^{N} A^k B \int_0^T e^{-\rho(T-\tau)}(T-\tau)^{\alpha-1} \frac{(T-\tau)^\alpha}{\Gamma(\alpha k+\alpha)} u(\tau)d\tau. \end{aligned}$$

In the preceding series, each term is expressed as a linear combination of the columns of B, AB, A^2B, …, $A^N B$. Each of these matrices can be represented as a linear combination of B, AB, A^2B, …, $A^{n-1}B$. Therefore, the vector

$$\sum_{k=0}^{N} A^k B \int_0^T e^{-\rho(T-\tau)}(T-\tau)^{\alpha-1} \frac{(T-\tau)^\alpha}{\Gamma(\alpha k+\alpha)} u(\tau)d\tau$$

is also a linear combination of the columns of B, AB, A^2B, …, $A^{n-1}B$, implying that it lies within the range space of the Kalman matrix K. Consequently, we obtain that $\mathcal{R}(K) = \mathbb{R}^n$. □

Remark 3.2. The condition $\mathcal{R}(K) = \mathbb{R}^n$ is necessary for the controllability of the linear fractional system (5): the Kalman matrix must have full rank, ensuring that we can only reach states within the range of the Kalman matrix.

3.2. Observability analysis

Now, we shall study the observability of the tempered fractional linear system (5) given by

$$\begin{cases} {}^T D_0^{\alpha,\rho} y(t) = Ay(t) + Bu(t), & t \in [0,T], \\ y(0) = y_0, & y_0 \in \mathbb{R}^n, \end{cases} \tag{21}$$

and augmented with the output function

$$z(t) = Cy(t) + Du(t), \tag{22}$$

where $z(t) \in \mathbb{R}^p$, $C \in \mathbb{R}^{p\times n}$ is the observation matrix, and C and $D \in \mathbb{R}^{m\times n}$ are assumed to be known constant matrices.

Definition 3.3. System (21)–(22) is said to be state observable on $[0,t_f]$, whenever any initial condition $y(0) = y_0 \in \mathbb{R}^n$ is distinctively determined via the associated system input $u(t)$ and output $y(t)$, for any $t \in [0,t_f]$, $t_f \in [0,T]$.

We now derive necessary and sufficient conditions for the observability of system (21)–(22).

Theorem 3.4. *System (21)–(22) is observable on $[0,t_f]$ if, and only if, the observability Gramian matrix*

$$W_o\left[0,t_f\right] := \int_0^{t_f} \mathrm{e}^{-\rho t} E_\alpha^*\left(At^\alpha\right) C^* C\, \mathrm{e}^{-\rho t} E_\alpha\left(At^\alpha\right) \mathrm{d}t$$

is non-singular for some $t_f > 0$.

Proof. The solution of system (21) is given by

$$y(t) = \mathrm{e}^{-\rho t} E_\alpha\left(At^\alpha\right) y_0 + \int_0^t \mathrm{e}^{-\rho(t-\tau)}(t-\tau)^{\alpha-1} E_{\alpha,\alpha}\left(A(t-\tau)^\alpha\right) Bu(\tau)\mathrm{d}\tau. \tag{23}$$

Combining formula (23) and the output (22), yields

$$y(t) = \mathrm{e}^{-\rho t} CE_\alpha\left(At^\alpha\right) y_0 + \int_0^t \mathrm{e}^{-\rho(t-\tau)}(t-\tau)^{\alpha-1} CE_{\alpha,\alpha}\left(A(t-\tau)^\alpha\right) Bu(\tau)\mathrm{d}\tau + Du(t).$$

Let us consider

$$\bar{y}(t) = z(t) - \int_0^t \mathrm{e}^{-\rho(t-\tau)}(t-\tau)^{\alpha-1} CE_{\alpha,\alpha}\left(A(t-\tau)^\alpha\right) Bu(\tau)\mathrm{d}\tau - Du(t).$$

Thus, one has

$$\bar{y}(t) = \mathrm{e}^{-\rho t} CE_\alpha\left(At^\alpha\right) y_0.$$

The observability of the system (21)–(22) is equivalent to the assessment of y_0 from $\bar{y}(t)$, which, in turn, since $\bar{y}(t)$ and y_0 are arbitrary, is equivalent to the evaluation of y_0 from $y(t)$, specified by

$$y(t) = \mathrm{e}^{-\rho t} CE_\alpha\left(At^\alpha\right) y_0,$$

where $u(t) = 0$. The observability Gramian matrix $W_o^{-1}\left[0,t_f\right]$ is well defined provided $W_o\left[0,t_f\right]$ is singular. Then, one gets

$$\begin{aligned} W_o^{-1}\left[0,t_f\right] &\int_0^{t_f} \mathrm{e}^{-\rho t} E_\alpha^*\left(At^\alpha\right) C^* y(t)\mathrm{d}t \\ &= W_o^{-1}\left[0,t_f\right] \int_0^{t_f} \mathrm{e}^{-\rho t} E_\alpha^*\left(At^\alpha\right) C^*\, \mathrm{e}^{-\rho t} CE_\alpha\left(At^\alpha\right) y_0 \mathrm{d}t \\ &= W_o^{-1}\left[0,t_f\right] \int_0^{t_f} \mathrm{e}^{-\rho t} E_\alpha^*\left(At^\alpha\right) C^* C\, \mathrm{e}^{-\rho t} E_\alpha\left(At^\alpha\right) y_0 \mathrm{d}t \\ &= W_o^{-1}\left[0,t_f\right] W_o\left[0,t_f\right] y_0 \\ &= y_0 \end{aligned}$$

for arbitrary $y(t)$ and $t_f > 0$. Therefore,

$$W_o^{-1}\left[0,t_f\right] \int_0^{t_f} \mathrm{e}^{-\rho t} E_\alpha^*\left(At^\alpha\right) C^* y(t)\mathrm{d}t = y_0. \tag{24}$$

One has that the left side of (24) is based on $y(t) \in \left[0,t_f\right]$, and (24) is a linear algebraic equation of y_0. Using the fact that $W_o^{-1}\left[0,t_f\right]$ exists, it follows that the initial condition $y(0) = y_0$ is distinctively determined by means of the associated system output $y(t)$ for $t \in \left[0,t_f\right]$. On the other hand, if $W_o\left[0,t_f\right]$ is singular for some $t_f > 0$, then there exists a vector $y_\alpha \neq 0$ fulfilling

$$y_\alpha^* W_o\left[0,t_f\right] y_\alpha = 0.$$

Taking $y_\alpha = y_0$, one has

$$y_\alpha^* \int_0^{t_f} E_\alpha^* (At^\alpha) C^* \mathrm{e}^{-\rho t} \mathrm{e}^{-\rho t} C E_\alpha (At^\alpha) y_\alpha \mathrm{d}t = 0.$$

This implies that

$$\int_0^{t_f} \|y(t)\|^2 \mathrm{d}t = 0.$$

Hence,

$$y(t) = \mathrm{e}^{-\rho t} C E_\alpha (At^\alpha) y_0$$

and, since the system is observable, it implies that $y_0 = 0$, which is a contradiction. We conclude that $W_o [0, t_f]$ is non-singular. □

We end Section 3.2 by proving the Kalman condition for the observability of system (21)–(22).

Theorem 3.5. *System (21)–(22) is observable on $[0, t_f]$ if, and only if,*

$$\mathit{rank}\ Q_o = \mathit{rank} \begin{pmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{pmatrix} = n.$$

Proof. From the proof of Theorem 3.4, one has

$$y(t) = \mathrm{e}^{-\rho t} C E_\alpha (At^\alpha) y_0,$$

where y_0 is uniquely determined by $y(t)$ if, and only if, $C E_\alpha (At^\alpha)$ is non-singular. Using the Cayley–Hamilton theorem, we obtain that

$$\begin{aligned} C E_\alpha (At^\alpha) &= C \sum_{j=0}^{\infty} \frac{t^{\alpha j}}{\Gamma(\alpha j + j)} A^j \\ &= C \sum_{j=0}^{n-1} \lambda_j t A^j \\ &= \sum_{j=0}^{n-1} \lambda_j t C A^j, \end{aligned}$$

where λ_j is a polynomial in $t - a$. The above formula may be written in matrix form as follows:

$$C E_\alpha (At^\alpha) = \left(\lambda_0 t, \lambda_1 t, \dots, \lambda_{n-1} t\right) \begin{pmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{pmatrix}.$$

Therefore, $C E_\alpha (At^\alpha)$ is non-singular if, and only if, rank $\begin{pmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{pmatrix} = n$. Then, we conclude that system (21)–(22) is observable on $[0, t_f]$ if, and only if, rank $\mathrm{Q}_o = n$. □

4. Applications

In this section, we present two examples to illustrate our main theoretical results.

4.1. Example 1: The fractional-order Chua's circuit

Chua's circuit, commonly referred as Chua's oscillator, is a basic electronic circuit that demonstrates nonlinear dynamic behaviors, such as bifurcation and chaos. This circuit consists of several components, including resistors, capacitors, and operational amplifiers. The fractional-order Chua's system with incommensurate parameters and external disturbance, as described in [24], is defined as follows:

$$\begin{cases} {}^T D_0^{\alpha,\rho} y_1(t) = \delta \left(y_2(t) - y_1(t) - m_1 y_1(t) - \frac{1}{2} \left(m_0 - m_1\right) \times (|y_1(t) + 1| - |y_1(t) - 1|) \right), \\ {}^T D_0^{\alpha,\rho} y_2(t) = y_1(t) - y_2(t) + y_3(t), \\ {}^T D_0^{\alpha,\rho} y_3(t) = -\beta y_2(t) - \gamma y_3(t) + u(t), \\ y_1(0) = y_{10}, \quad y_2(0) = y_{20}, \quad y_3(0) = y_{30}, \end{cases} \tag{25}$$

where $y_3(t)$ is the current through the inductance, and $y_1(t)$ and $y_2(t)$ are the voltages across the capacitors. In [25], when the tempered derivative in (25) is replaced with the generalized Caputo proportional fractional-order derivative, to consider the system as a linear system, the authors evaluated $\frac{1}{2}\left(m_0 - m_1\right) \times (|y_1(t)+1| - |y_1(t)-1|)$ for $y_1(t) \geq 1$, $-1 < y_1(t) < 1$ and $y_1(t) \leq -1$. However, this approach might appear unusual, as the term $\frac{1}{2}\left(m_0 - m_1\right) \times (|y_1(t)+1| - |y_1(t)-1|)$ is a component of the state solution, factored into the determination of the solution, and is no longer a constant to be compared with the fixed values -1 and 1. To remove any doubt, we linearize the system (as is customary) around the origin, yielding the following system:

$$\begin{cases} {}^{T}D_0^{\alpha,\rho} y_1(t) = \delta\left(y_2(t) - y_1(t) - m_0 y_1(t)\right), \\ {}^{T}D_0^{\alpha,\rho} y_2(t) = y_1(t) - y_2(t) + y_3(t), \\ {}^{T}D_0^{\alpha,\rho} y_3(t) = -\beta y_2(t) - \gamma y_3(t) + u(t), \\ y_1(0) = y_{10}, \quad y_2(0) = y_{20}, \quad y_3(0) = y_{30}. \end{cases} \tag{26}$$

The linearized system (26) can be expressed in the abstract form of (5) as

$$\begin{cases} {}^{T}D_0^{\alpha,\rho} y(t) = Ay(t) + Bu(t), \\ y(t) = y_0, \end{cases} \tag{27}$$

where

$$y(t) = \begin{pmatrix} y_1(t) \\ y_2(t) \\ y_3(t) \end{pmatrix}, \quad y_0 = \begin{pmatrix} y_{10} \\ y_{20} \\ y_{30} \end{pmatrix}, \quad A = \begin{pmatrix} -\delta(1+m_0) & \delta & 0 \\ 1 & -1 & 1 \\ 0 & -\beta & -\gamma \end{pmatrix} \text{ and } B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

The controllable matrix has the form

$$\begin{bmatrix} 0 & 0 & \delta \\ 0 & 1 & -1-\gamma \\ 1 & -\gamma & -\beta+\gamma^2 \end{bmatrix}. \tag{28}$$

Moreover, a simple calculation shows that $\det\left[B, AB, A^2B\right] = \delta \neq 0$, which implies that $rank\left[B, AB, A^2B\right] = 3$. Therefore, from Theorem 3.2, the tempered fractional Chua's circuit is controllable.

Now, we give some numerical simulations for the controllability of system (26) over the interval $[0, 1.5]$ in the case where $\rho = 0.5$ and $\alpha = 0.7$, and by taking the parameters $\delta = 2$, $\beta = 0.5$, $\gamma = -1$, and $m_0 = 3$. According to system (27), one has

$$A = \begin{pmatrix} -8 & 2 & 0 \\ 1 & -1 & 1 \\ 0 & -0.5 & 1 \end{pmatrix}.$$

By virtue of the controllability Gramian matrix formula (17), we have

$$W_c[0, 1.5] = \int_0^{1.5} \mathrm{e}^{-(1.5-\tau)}(1.5-\tau)^{-0.6} E_{0.7,0.7}\left((1.5-\tau)^{0.7}A\right) BB^* E_{\alpha,\alpha}\left((1.5-\tau)^{0.7}A^*\right) \mathrm{d}\tau. \tag{29}$$

Using the Mittag-Leffler function of two parameters (4), we obtain that

$$W_c[0, 1.5] = \begin{pmatrix} 17.4120 & 9.7422 & 10.0512 \\ 9.7422 & 2.8135 & 24.3021 \\ 10.0512 & 24.3021 & 25.3369 \end{pmatrix},$$

which is non-singular for $T = 1.5$. Then, from Theorem 3.2, we conclude that system (26) with $\rho = 0.5$ is controllable on $[0\ 1.5]$. We shall use the control $u(t)$ defined by formula (19) as follows:

$$\begin{aligned} u(t) &= -(1.5-t)^{-0.3}\, \mathrm{e}^{-0.5(1.5-t)}(B)^* E_{0.7,0.7}\left(A^*(1.5-t)^{0.7}\right) \\ &\quad \times W_c^{-1}[0, 1.5]\, \mathrm{e}^{-0.75} \times E_{0.7}\left(1.5^{0.7}A\right) y_0. \end{aligned} \tag{30}$$

This control can steer the system (26) from the initial state $y(0) = [2; 5; 3]$ to the desired state $y(1.5) = [1; -3.5; 3.5]$.

Fig. 1 presents the trajectories of system (26) without the control $u(t)$. We observe that there is no trajectory connecting the initial state $[2; 5; 3]$ to the final desired state $[1; -3.5; 3.5]$. In Fig. 2, we examine the behavior of system (26) using the steering control $u(t)$. It shows that the state of system (26) moves from its initial state to the desired final state $[1; -3.5; 3.5]$ over the interval $[0,\ 1.5]$. The evolution of the steering control function $u(t)$ is illustrated in Fig. 3.

Now, we present numerical simulations for the controllability of system (26) with $\rho = 1$ on the interval $[0, 1.5]$. We again take the parameter values $\delta = 2$, $\beta = 0.5$, $\gamma = -1$, and $m_0 = 3$.

From the controllability Gramian matrix formula (17), we have

$$W_c[0, 1.5] = \int_0^{1.5} \mathrm{e}^{-2(1.5-\tau)}(1.5-\tau)^{0.6} E_{0.7,0.7}\left((1.5-\tau)^{0.7}A\right) BB^* E_{0.7,0.7}\left((1.5-\tau)^{0.7}A^*\right) \mathrm{d}\tau.$$

Using the Mittag-Leffler function of two parameters (4) and performing some numerical calculations, we get

$$W_c[0, 1.5] = \begin{pmatrix} 18.2342 & 10.2913 & 10.4490 \\ 10.2913 & 3.7710 & 24.1524 \\ 10.4490 & 24.1524 & 27.5196 \end{pmatrix},$$

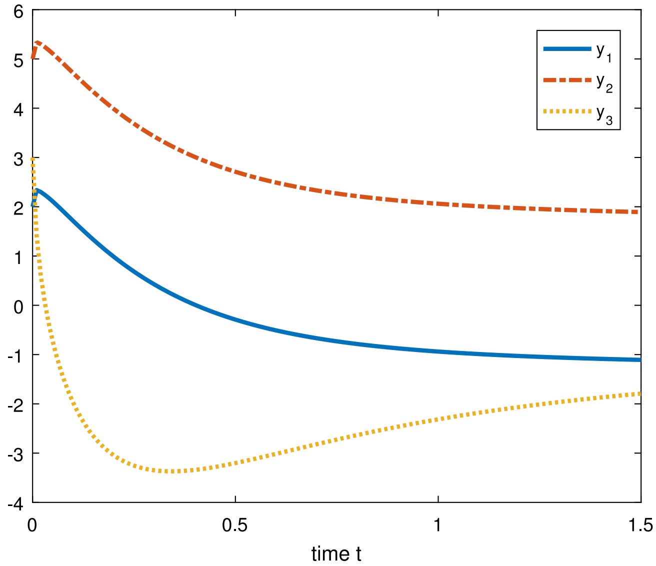

Fig. 1. The system (26) evolution on the interval [0 1.5] with $\rho = 0.5$ and $u(t) = 0$.

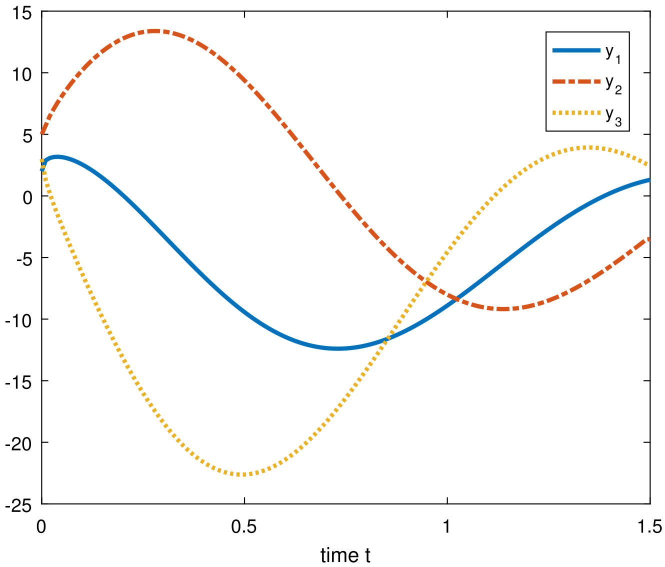

Fig. 2. The evolution of the controlled system (26) with $\rho = 0.5$ on the interval [0 1.5].

which is non-singular for $T = 1.5$. Then, by virtue of Theorem 3.2, we deduce that system (26) with $\rho = 1$ is controllable on [0 1.5]. We use the control $u(t)$ defined by formula (19) as follows:

$$u(t) = -(1.5-t)^{-0.3}\, e^{-1.5+t}(B)^* E_{0.7,0.7}\left(A^*(1.5-t)^{0.7}\right) \times W_c^{-1}[0,1.5]\, e^{-T} \times E_{0.7}\left(1.5^{0.7}A\right) y_0 \tag{31}$$

that allows us to steer the system (26) from the initial state $y(0) = [4; 2; -3]$ to the final state $y(1.5) = [5; 9; 15]$.

In Fig. 4, we examine the solution of system (26) without the control $u(t)$. We observe that there is no trajectory connecting the initial state $[4; 2; -3]$ and the final state $[5; 9; 15]$. In Fig. 5, we illustrate the behavior of system (26) using the steering control $u(t)$. It shows that the state of system (26) moves from its initial state to the final state $[5; 9; 15]$ over the interval $[0,\ 1.5]$. The steering control function $u(t)$ is shown in Fig. 6.

4.2. Example 2: Chua–Hartley's oscillator

The Chua–Hartley system differs from the conventional Chua system in such a way that it substitutes the piecewise-linear nonlinearity with an appropriate cubic nonlinearity, resulting in a very similar behavior. Here, we replace the integer derivatives

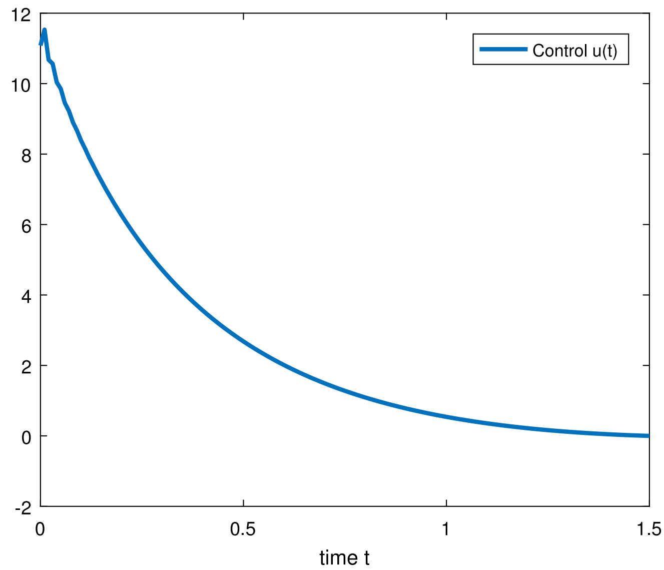

Fig. 3. The steering control (30) on [0 1.5] in the case $\rho = 0.5$.

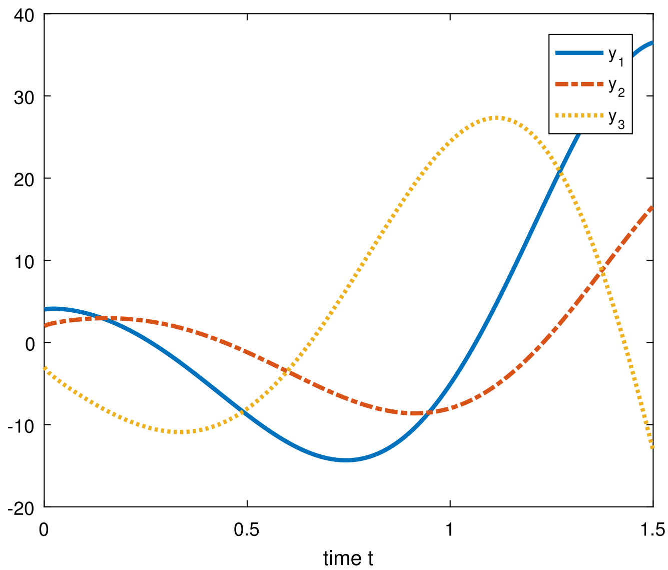

Fig. 4. The system (26) evolution on the interval [0 1.5] with $\rho = 1$ and $u(t) = 0$.

on the left side of the Chua–Hartley differential system [26] by tempered fractional derivatives, as follows:

$$\begin{cases} {}_0{}^{T}D_0^{\alpha,\rho} y_1(t) = \delta\left(y_2(t) + \dfrac{y_1(t) - 2y_1^3(t)}{7}\right), \\ {}^{T}D_0^{\alpha,\rho} y_2(t) = y_1(t) - y_2(t) + y_3(t) + u(t), \\ {}^{T}D_0^{\alpha,\rho} y_3(t) = -\beta y_2(t) = -\dfrac{100}{7} y_2(t). \end{cases} \tag{32}$$

Also, we consider that the system is augmented with the output functions

$$\begin{cases} z_1(t) = y_1(t) + y_3(t), \\ z_2(t) = 2y_1(t) - y_2(t). \end{cases} \tag{33}$$

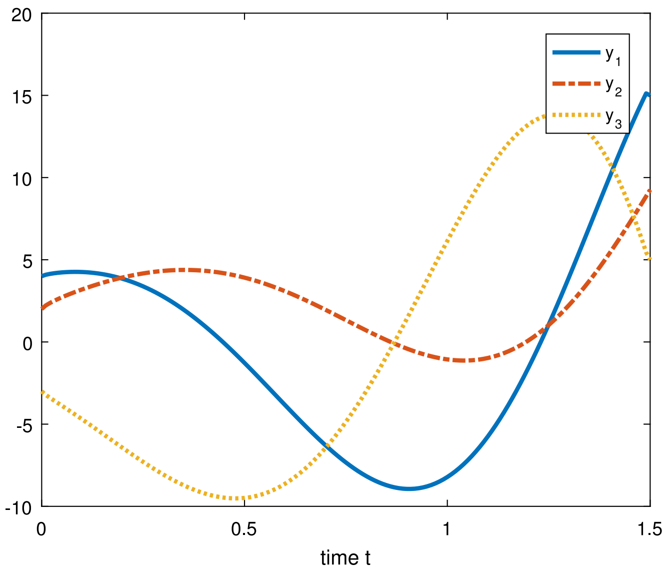

Fig. 5. The evolution of the controlled system (26) with $\rho = 1$ on the interval [0 1.5].

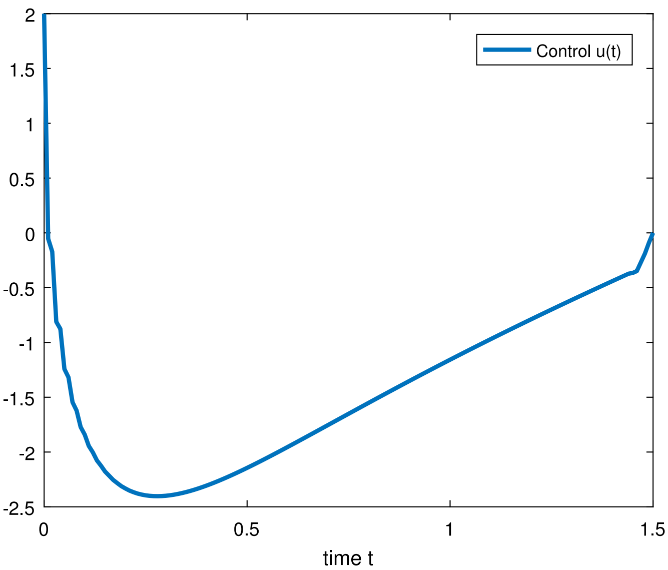

Fig. 6. The steering control (31) on [0 1.5] in the case $\rho = 1$.

Linearizing the system (32) around the origin, one obtains that

$$\begin{cases} {}_0^T D_0^{\alpha,\rho} y_1(t) = \delta y_2(t) + \dfrac{\delta}{7} y_1(t), \\ {}^T D_0^{\alpha,\rho} y_2(t) = y_1(t) - y_2(t) + y_3(t) + u(t), \\ {}^T D_0^{\alpha,\rho} y_3(t) = -\dfrac{100}{7} y_2(t). \end{cases} \tag{34}$$

According to (21)–(22), the linearized system (34)–(33) can be expressed in the following abstract form:

$$\begin{cases} {}^T D_0^{\alpha,\rho} y(t) = A y(t) + B u(t), \\ y(t) = y_0, \end{cases} \tag{35}$$

and

$$z(t) = C y(t), \tag{36}$$

where

$$y(t)=\begin{pmatrix} y_1(t)\\ y_2(t)\\ y_3(t)\end{pmatrix},\ A=\begin{pmatrix} \frac{\delta}{7} & \delta & 0\\ 1 & -1 & 1\\ 0 & -\frac{100}{7} & 0\end{pmatrix},\ B=\begin{pmatrix}0\\1\\0\end{pmatrix}, z(t)=\begin{pmatrix} z_1(t)\\ z_2(t)\end{pmatrix} \text{ and } C=\begin{pmatrix} 1 & 0 & 1\\ 2 & -1 & 0\end{pmatrix}.$$

Simple calculations show that

$$\begin{aligned} rank\ Q_o &= rank \begin{pmatrix} C\\ CA\\ CA^2\end{pmatrix}\\ &= rank \begin{pmatrix} 1 & 0 & 1\\ 2 & -1 & 0\\ \frac{\delta}{7} & \delta-\frac{100}{7} & 0\\ \frac{2\delta}{7}-1 & 2\delta+1 & -1\\ \frac{2\delta^2}{49}+2\delta-\frac{100}{7} & \frac{2\delta^2}{7}-2\delta+\frac{100}{7} & 2\delta-\frac{100}{7}\\ \frac{2\delta^2}{49}+\frac{10\delta}{7}+1 & \frac{2\delta^2}{7}-3\delta+\frac{93}{7} & 2\delta+1\end{pmatrix}\\ &=3. \end{aligned}$$

Therefore, from Theorem 3.5, the tempered linearized Chua–Hartley oscillator is observable.

5. Conclusion

In this paper, we studied the well-posedness, controllability, and observability of tempered fractional differential systems. The main contribution is the proof of necessary and sufficient Gramian matrix criteria and rank conditions for both the controllability and observability of linear tempered fractional differential systems. To illustrate the theoretical results, we provided examples using fractional Chua's circuit and Chua–Hartley oscillator models.

Controllability results for nonlinear fractional-order dynamical systems can be found in [27,28]. In the future, we plan to investigate the controllability and observability problems for nonlinear tempered fractional differential systems.

CRediT authorship contribution statement

Ilyasse Lamrani: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization, Software, Visualization. **Hanaa Zitane:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization, Software, Visualization. **Delfim F.M. Torres:** Writing – review & editing, Writing – original draft, Validation, Supervision, Investigation, Funding acquisition, Conceptualization, Formal analysis, Methodology.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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